

TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

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VERSION **v. 1.1**

RELATED SUPPORT

- **TEMPLATE GUIDE Key Project Information & VPA Design Document v.1.1**

This document contains the following Sections

Key Project Information

Section A –Description of project

Section B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

Section C – Duration and crediting period

Section D – Summary of Safeguarding Principles and Gender Sensitive Assessment

Section E– Summary of Local stakeholder consultation

Appendix 1 – Safeguarding Principles Assessment (mandatory)

Appendix 2 - Contact information of VPA Implementer (mandatory)

Appendix 3 - Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of Project	11440
Title of Project	GS10884 - KOKO Kenya - Ethanol Cookstoves Program - CPA-0002
Time of First Submission Date	18/11/2021
Date of Design Certification	25/08/2022
Version number of the VPA-DD	Version 5.0
Completion date of version	25/09/2022
Coordinating/managing entity	KOKO Networks Limited
VPA Implementer (s)	KOKO Networks Limited
Project Participants and any communities involved	None
Host Country (ies)	Kenya
GS ID and Title of applicable Design Certified VPA	N/A
GS ID and Title of applicable Performance Certified VPA	N/A
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☒ Small Scale ☐ Large Scale

Other Requirements applied	Programme of Activity Requirements, Version 1.2 Community Services Activity Requirements, Version 1.2
Methodology (ies) applied and version number	SDG 13: GS labeling of UNFCCC CERs. CERs issued with use of AMS-I.E.: Switch from non-renewable biomass for thermal applications by the user, Version 09.0
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6)	Estimated Annual Average	Units or Products
SDG 13 Climate Action (mandatory)	GHG Emissions Reductions	1,081,893 tCO ₂ /year	(GS-CERs/VERs)
SDG 3 Ensure healthy lives and promote well-being for all at all ages	Number of KOKO Users /households experiencing reduction in the indoor air pollution in their kitchen	149,250	Users
SDG 5 Gender Equality	Number of women in managerial positions	15	Women employees (Apportioned for CPA-2)
SDG 7 Access to Clean Energy	Proportion of population or number of households with primary reliance on clean fuels and technology	199,000	KOKO Cooker Beneficiaries

Section A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

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The VPA involves distribution of bio-ethanol fuel cookstoves and promote its use for cooking purposes in households in Kenya.

KOKO enables the distribution of high efficiency ethanol cook stoves and last-mile delivery of bio-ethanol fuel to the target customers using cloud systems and internet-of-things (IoT) technology to allow low-cost affordable access to clean fuel. The CPA initially planned to distribute approximately two hundred thousand (200,000) cookstoves across Kenya in phase 1 and depending on further demand, it was expected to continue to include more cookstoves in Kenya. The CPA can include unlimited cookstoves during its crediting period as it complies with the requirements of micro-scale CDM units. A relevant deviation (DEV_247) from small-scale threshold is also approved by Gold Standard to allow unlimited cookstoves under this CPA. The VPA, after completing its phase 1, expanded the scope and has distributed 209,817 cookstoves till 30th November 2021 (time of 1st submission to GS). The baseline scenario, i.e. in the absence of the project activity, would be the use of predominantly non-renewable woody biomass and traditional stoves for meeting similar thermal energy cooking needs. In absence of implementation of the proposed VPA, the households would have continued the existing pre-project scenario, namely to use unimproved cookstoves consuming in most cases woody biomass, consisting of firewood and charcoal.

A.1.1. Eligibility of the project under approved PoA

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The eligibility of technology under section 3.1.1 of GS4GG Principles & Requirements version 1.2 and Community Services Activity Requirements version 1.2 /17/ is as follows:

Eligibility Criteria	Justification from PP
3.1.1 (a) Types of Project Eligible projects shall include physical action/implementation on the ground. Pre- identified eligible project types are	The PoA involves implementation on the ground i.e. distribution of bioethanol stoves which comes under section 3.1.1 (b) of the approved Gold Standard Activity Requirement -Community Services Activity Requirements Version 1.2

identified in the Eligibility Principles and Requirements section.	Published October 2019. Therefore, there is approved Gold Standard Activity Requirements and thus the project is automatically eligible.
3.1.1 (b) Location of Project Projects may be located in any part of the world.	PoA is located in Kenya.
3.1.1 (c) Project Area, Project Boundary and Scale The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects).	Physical/ Geographical boundary of the PoA is Kenya. The same has been defined in detail in section A.3 of the PoA DD. Scale: This is a CDM-registered PoA and has demonstrated that the included CPAs will only consist of micro-scale CDM units. As per CDM guidelines, the CPAs consisting of microscale CDM units are not required to demonstrate compliance with the micro-scale CDM thresholds at the aggregate level of the CPA. The requirement of Annex B Para 5.1.2. and 5.1.3. GHG Emission Reductions and Sequestration Product Requirements mandates the transitioning PoAs to define the scale as per CDM guidance. Hence, the CME has submitted a deviation from the small-scale threshold limits as defined in section 9.1.2 of the GHG Product Requirements Standard and request the Gold Standard to conduct design review of the project by evaluating it against the guidelines of Annex B of the GHG Product requirements. The review was approved on 27th April 2022. The approved deviation (DEV_247) has been submitted along with the design review request. Double counting: The bioethanol stoves will bear a unique serial number and project database will record user location details. Unique serial number and user location details will identify the project devices individually and will separate it from other programs, if any in the present or in the future.
3.1.1 (d) Host Country Requirements Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	Projects is in compliance with applicable Host Country's legal, environmental, ecological and social regulations.
3.1.1 (e) Contact Details	Name and Contact details of the project developer have been provided in the Appendix 1 of the PoA DD.

As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	Legal registration details has been provided in the Appendix 1 of the PoA DD. Company Registration certificate has been provided.
3.1.1 (f) Legal Ownership Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	There is a provision in the project device purchase agreement with manufacturer that emission reductions achieved by the project technology are owned by the project developer. Provision in the project device sales agreement (retailer) that emission reductions achieved by the project technology are owned by the project developer. The CME has signed the sales agreement for end users including the provision that emission reductions achieved by the project technology are owned by the CME.
3.1.1 (g) Other Rights As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to	The PoA does not involve any kind of production activity or use of natural resources. Therefore, legal rights and/or permissions concerning changes in use of other resources required to service the project is not required.

further Project implementation in affected areas.	
3.1.1 (h) Official Development Assistance (ODA) Declaration All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.	The project developer has submitted Official Development Assistance (ODA) Declaration.

General Eligibility Criteria as per section 3 of Community Services Activity Requirements version 1.2:

Eligibility Criteria	Justification from PP
3.1.1 Types of project – Pre-identified CSA project types are noted below. (a)Renewable energy Project Developers may submit new project types to Gold Standard for approval following the Principles & Requirements. (b) End-use energy efficiency: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end-user of the products and services are clearly identified and when the physical intervention is	Applicable criteria (b) End-use energy efficiency, which is an eligible project type under section 3.1.1(b) of the Community Services Activity Requirements PoA involves dissemination of fuel-efficient stoves that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products.

required at the user end. For example, efficient cooking, heating, lighting, etc.	
3.1.2 Project Area, Boundary and Scale Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies or Product Requirements. The definition of scale is the same for all Projects, except Microscale which is defined as: (a) CSA Project issuing emission reductions less than or equal to 10,000 tCO₂eq per annum (b) CSA Project seeking any Gold Standard Certified Impact or Product other than emission reductions and meeting one of the following criteria: (a) Installed capacity less than equal to 2 MWe₁ /6 MWth that employs renewable energy as the primary technology (b) Energy savings at a scale of no more than 20 GWh per year where energy efficiency is the primary activity Achieve GHG emissions reductions at a scale of no more than 20,000 tCO₂eq per annum where project activity type is not included in the above two criteria.	The PoA has applied UNFCCC methodology- AMS I.E V09 "Switch from non-renewable biomass for thermal applications by the user" Project boundary- Kenya Project scale- Small scale. The scale of the project defines by the CDM guidance and the compliance of scale which meets the requirement of Annex B criteria 5.1.2. and 5.1.3. GHG Emission Reductions and Sequestration Product Requirements which means that the project activity will remain same as per the CDM registered PoA. The CME has submitted a deviation from the small-scale threshold limits as defined in section 9.1.2 of the GHG Product Requirements Standard and request the Gold Standard to conduct design review of the project by evaluating it against the guidelines of Annex B of the GHG Product requirements. The review was approved on 27th April 2022. The approved deviation (DEV_247) has been submitted on GS portal.

(c) For the purpose of applying UNFCCC methodologies for quantification of GHG reductions, 'small scale' is defined as in CDM Modalities and Procedures for three projects types; Renewable Energy, Energy Efficiency and Others.	
3.1.3 Certain Impact Quantification methodologies allow projects to account Suppressed Demand scenario when establishing a baseline. In such cases, the application of Suppressed Demand baseline is limited to Small Scale and Microscale Projects. Where a Suppressed Demand baseline is applied, it is not possible to 'stack' Gold Standard Certified Impact Statements or Products as the definition of the baseline may be contradictory.	The concept of suppressed demand has not been used in establishing the baseline.
3.1.4 Legal ownership: (a) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the	Legal ownership: (a)End user sales agreement with a clear description of the ownership of the Products that are generated under Gold Standard Certification. (b)Transfer and ownership of the Product was discussed during the local stakeholder meeting.

proofs that end-users are aware of and willing to give up their rights on Products shall be provided. (b) The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	
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S.No.	Eligibility Criterion	Description/Required condition	Means of Verification/ Supporting evidence for inclusion
1	Ref: Project Standard for PoA Version 2.0 Para.120(a): The geographical boundary of the CPA including any time-induced boundary consistent with the geographical boundary set in the PoA	The PoA boundary corresponds to the boundaries of Kenya. Each CPA/VPA will be located within Kenya.	Customer Database with Neighborhood
2	Ref: Project Standard for PoA Version 2.0 Para.120(b): Conditions that avoid double counting of emission reductions like unique identification of product and end-user locations	A unique numbering system for cookstoves will be applied in each CPA/VPA, assigning a unique number to each cookstove and allowing to clearly identify for each cookstove to which CPA/VPA it belongs. Before handing over the cookstove to the end-user, the complete information of the end-user is electronically registered and stored on the cloud server and the cookstove serial number is assigned to the registered user.	Customer Database with Unique Serial Number of cookstove

3	Ref: Project Standard for PoA Version 2.0 Para.120(c): Conditions to confirm that CPAs are neither registered as CDM project activities, included in another registered PoAs, nor the project activities that have been Deregistered	The CPA/VPA is exclusively bound to the PoA. Confirmation that the programme activity has not been and will not be registered either as a single CDM project activity or as a CPA under another PoA or under a project activity which has been registered.	A statement is provided by the CPA/VPA implementer that the CPA/VPA is not part of any other carbon offset project activity.
4	Ref: Project Standard for PoA Version 2.0 Para.120(d): The specifications of technology/measure including the level and type of service, performance specifications including compliance with testing/certifications	1. Technology - The PoA will promote dissemination of portable bio-ethanol cookstoves in Kenya 2. Key Specifications: • Thermal Output: Between 1.4 kW and 68.5 kW • Thermal Efficiency: More than 45% • Fuel: Bioethanol The Cookstoves, to be distributed under the CPA/VPA, are designed and constructed in accordance with the national safety and performance regulation on bio-ethanol cookstoves of Kenya – “Ethanol Fuelled Cooking Appliances Specifications”.	Technical Specification Details to be made available by the CPA/VPA to demonstrate compliance with the eligibility conditions.
5	Ref: Project Standard for PoA Version 2.0 Para.120(e): Conditions to check the start date of the CPA through documentary evidence	Date on which first cookstove is sold/installed under the CPA/VPA. The start date of any proposed CPA/VPA will be on or after the start date of the proposed PoA.	End user agreement / voucher / installation report etc for the first cookstove sold/installed under the CPA/VPA.
6	Ref: Project Standard for PoA Version 2.0 Para.120(f): Conditions that ensure compliance with applicability and other requirements of the methodology	CPA must follow AMS-I.E. Version 09.0. The applicability of methodology at CPA level is demonstrated in section B.2 below. Technology related requirements have been specified in criteria #3 above.	CPA-DD/VPA-DD applying methodology AMS-I.E. Version 09.0.

7	Ref: Project Standard for PoA Version 2.0 Para.120(g): The conditions that ensure that the CPA meets the requirements pertaining to the demonstration of additionality	1. The cookstove shall be used in household or SME. The unit capacity of the cookstove shall be less than 68.5 kW.	1. The verification of the end-user will be done at the time of sale of cookstove. The registration details will be available as proof of verification. 2. The specifications of the cook stove models will be approved by the CME before distribution. Proof: Cookstove Specification Details
8	Ref: Project Standard for PoA Version 2.0 Para.120(h): The PoA-specific requirements stipulated by the CME including any conditions related to undertaking local stakeholder consultations and environmental impact analysis	The local stakeholder consultation is conducted at the PoA level (Section F of the PoA-DD). An environmental impact analysis is not required (section E of the PoA-DD)	Not Applicable
9	Ref: Project Standard for PoA Version 2.0 Para.120(i): Conditions to provide an affirmation that funding from Annex I Parties, if any, does not result in a diversion of official development assistance	Affirmation that funding from Annex I Parties, if any, does not result in a diversion of official development assistance	Please refer Appendix 2. Declaration from CME that no funds for official development assistance will be used for program implementation.
10	Ref: Project Standard for PoA Version 2.0 Para.120(j): Where applicable, target group (e.g. domestic/commercial/industrial, rural/urban, grid-connected/off-grid) and distribution mechanisms (e.g. direct installation)	Target Group: Households Cookstove Distribution Mechanism: Direct Distribution through KOKOpoints	Customer Database and associated KOKO Agent details

11	Ref: Project Standard for PoA Version 2.0 Para.120(k): Where applicable, the conditions related to sampling requirements for the PoA in accordance with the "Standard for sampling and surveys for CDM project activities and programme of activities"	CPAs/VPAs under the program will adhere to all requirements as mentioned in Standard: "Sampling and surveys for CDM project activities and programme of activities"	CPAs/VPAs will follow monitoring plan described in CPA-DD/VPA-DD.
12	Ref: Project Standard for PoA Version 2.0 Para.120(l): Where applicable, the conditions that ensure that every CPA (in aggregate if it comprises of independent sub units) meets the small-scale or microscale threshold and remains within those thresholds throughout the crediting period of the CPA	CME will ensure that all CPAs are solely composed of "microscale CDM units", hence the CME is not required to demonstrate compliance with the small-scale CDM thresholds at the aggregate level of the CPA. As a result, unlimited cookstove units can be included in a CPA under this PoA. In this regard, GS has also approved a deviation for this project – DEV_247 which exempts this VPA from demonstrating compliance to aggregate small-scale threshold.	Not Applicable
13	Ref: Project Standard for PoA Version 2.0 Para.120(m): Where applicable, the requirements for the debundling check, in case the CPAs belongs to small-scale or microscale project categories.	CME will ensure that all CPAs are solely composed of "microscale CDM units", hence the CME is not required to perform de-bundling check. Please refer the Deviation DEV_247 for further details.	Not Applicable
14	Approval of CPA by CME	The CME approves each CPA/VPA to be included into its registered PoA after detailed technical review.	Statement of CME giving approval for the CPA/VPA to be included into its registered PoA along with technical review report.

15	CER ownership	End users receiving cookstove under the specific CPA/VPA cede their rights to claim and own emission reductions under the Clean Development Mechanism of the UNFCCC to the CME of the PoA.	A default terms and conditions for end users including the provision that emission reductions generated by the stove are owned by the CME will be provided for each CPA/VPA
16	SDG Outcome Assessment	VPA shall demonstrate a clear, direct contribution to sustainable development, defined as making demonstrable, positive impacts on at least three SDGs, one of which must be SDG 13 (defined herein as Emissions Reductions or Removals and/or Adaptation to climate change)	CPA-0001 has demonstrated the positive impacts on SDG 3, SDG 5, SDG 7 and SDG 13.
17	Safeguarding Principles Assessment	Each CPA will conduct safety trainings to ensure safety of operational team/staff.	CPA conducts regular training and audit measures for the safety of the operating staff. Means of Verification: Training records
18	Retroactive VPA	VPAs with retroactive crediting period submit the required documents for Gold Standard Preliminary Review (time of first submission) within one year from the VPA start date. (This requirement is not applicable for CDM registered CPAs)	Not Applicable CPA-0002 is a CDM registered CPA, hence requirement is not applicable.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

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The transfer of impact/SDG contribution ownership throughout the investment chain is made transparent through digital contracts (or terms & conditions). Each KOKO customer waives their rights to SDG contributions at the purchase of the stove, transferring the ownership rights to the certified SDG contributions to KOKO Networks.

A.2. Location of project

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All counties, cities, towns, villages and communities within the geographical boundary of Republic of Kenya.



Fig 1: Political map of Kenya

Geo-Coordinates of Kenya

Southern-most point of Kenya	4°38'47.4"S 39°12'31.6"E
Western-most point of Kenya	0°06'58.9"N 33°57'35.3"E
Eastern-most point of Kenya	3°55'51.6"N 41°51'59.9"E
Northern-most point of Kenya	4°28'42.5"N 35°52'31.8"E

A.3. Technologies and/or measures

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Details of the Technology

Gold Standard

KOKO’s clean fuel technology, “KOKO Cooker” stove kit consists of a two-burner stove and a durable ‘smart’ canister equipped with an NFC chip that enables tracking of customer fuel purchases. To create affordable and reliable access to bio-ethanol as clean cooking fuel, KOKO has developed first-of-its-kind technology to lower distribution costs by launching a network of cloud-connected “KOKOpoint” fuel ATMs across Kenya, which enable customers to refill their smart canisters in local retail corner stores. Customers can buy their KOKO Cookers through ordering on the KOKOpoint tablet screen, via the myKOKO mobile app, or from a network of field sales promoters. Customers can also take advantage of the savings scheme launched on the KOKO platform, which enables low-income customers who have difficulty with the upfront cost of the KOKO Cooker to make progressive payments towards their stove. The technology will result in displacement of high GHG-emitting woody biomass as cooking fuel in the low-income households and commercial spaces of Kenya.

The technology will result in displacement of high GHG-emitting woody biomass as cooking fuel in the low-income households and commercial spaces of Kenya. In absence of implementation of the proposed CPA, the households would have continued to use unimproved cookstoves consuming woody biomass, consisting of firewood and charcoal, for cooking purpose which is also the existing pre-project scenario.

The procurement of the cookstoves for the PoA will comply with the national regulation on ethanol cookstoves – “Ethanol Fuelled Cooking Appliances Specifications”¹. As an illustration, the “KOKO Cooker” stove kit consisting of a two-burner stove and a durable ‘smart’ canister, with thermal efficiency of 60% and thermal output of 2.1 kW (against the requirement of 45% efficiency and 1.4 kW output) fully comply with the above standard. The detailed specification are as follows:

Description	2-burner Ethanol stove for domestic cooking purposes
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¹ Kenya Standard: Ethanol fueled cooking appliances – Specification (KS 2759:2018 – ICS 97.040.20)

Fuel Denatured	Ethanol Cooking Fuel (95% ethyl alcohol, 5% water; 22 MJ/kg lower heating value)
Firepower	2100 Watts on MAX flame setting; 1000 Watts on MEDIUM; 500 Watts on LOW
Overall Efficiency	60% (energy content of fuel transferred to contents of pot; as defined by ISO WBT methodology)
Fuel Capacity	2.4 Litres fuel capacity within stove
Burn Time	2.5 hours per burner on MAX setting (5 hours per burner on LOW setting) for fully fuelled stove
Expected Service Life	10 years with routine annual maintenance
Box Dimensions	90 cm (width) x 33 cm (depth) x 20 cm (height) ; with carry handle Box Weight (gross) 8 kg

The CME may launch more cookstove models with varying specifications as per market demand which are complying to the local regulations of Kenya. The CME will further ensure that the thermal output of all cookstove models distributed under the CPA shall not be higher than 1500 kW. Hence, all cookstoves under the CPA will qualify as “CDM micro unit” thereby demonstrating the additionality of the CPA. This is verifiable through the cookstove models distributed under the CPA.

Project Type

The Pre-identified Community Service Activity project type of the VPA is:

b) End-use energy efficiency: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end-user of the products and services are clearly identified and when the physical intervention is required at the user end. For example, efficient cooking, heating, lighting, etc.

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Claimed SDG contribution

The SDG impact analysis for the project is conducted using the SDG Impact Tool published by Gold Standard. The monitoring indicators are identified as per the results received by filling the SDG Impact Tool.

SDG 3: The use of the KOKO Cooker and fuel for the majority of the cooking significantly reduces the Household Air Pollution (HAP) in the home. HAP comprises levels of particulate matter (including fine particulate matter PM_{2.5}) and gases that can exceed air pollution exposures in even the most polluted urban settings. PM_{2.5} can penetrate deep into the lungs and trigger systemic inflammation and oxidative stress and considered the best indicator air pollutant to estimate the negative health impacts of air pollution. Exposure-response relationships between PM_{2.5} and risk of pneumonia, lung cancer, COPD, and cardiovascular outcomes estimate that exposure to HAP is detrimental to the health of infants, children, and adults. To minimize the disease burden of HAP, the WHO recommends achieving an interim indoor air quality target of 35 µg/m³ for annual mean PM_{2.5}. The WHO also advises against the indoor use of coal and kerosene. The KOKO technology is able to reduce more than 80% of the indoor air pollution and achieve the indoor air quality very close to the target figure. Since direct measurement of indoor air pollution in households is very expensive, it is not possible to include that in the project monitoring plan. However, the personal qualitative experience of users while cooking can be a good indicator of the change in indoor air pollution. Hence, the monitoring of indoor air pollution will be performed via user Monitoring Survey on the perception of users on reduction in indoor air pollution.

SDG 5: The project will create new employment opportunities for women across the value chain and will promote gender equality and equal opportunity for women to participate in economic activities. The KOKO technology will enable women empowerment placing women in management positions to take economic decisions and allowing the household women access to the digital world through KOKO app, reducing time spent by women on procuring fuel and cooking household meals and assuming ownership of the household asset. The following impact parameter will be monitored by the CME:

- a) Number of women in managerial positions

The use of the KOKO Cooker and fuel for the majority of cooking significantly reduces the burden on women in the home. Through its speed, cleanliness in use and convenience it will reduce the time pressure and enhance the status of cooking. KOKO Networks has also tried to promote equality in household responsibility via its advertisements and promotional campaigns.

SDG 7: The people of Kenya do not have access to clean cooking as the clean technology like LPG or electric cookers are capital intensive and unaffordable due to larger SKUs and high pricing. As a result, 90% of the households in Kenya are still not able to access clean fuel and rely on dirty non-renewable fuels for cooking. Bioethanol cooking is ultra-clean modern fuel, delivering no harmful emissions to users. KOKO fuel is highly affordable and can be accessed with as low as 30 KES as compared to LPG technology where smallest cylinder costs more than 600 KES. By switching from dirty fuels, customers are also saving up to 50% on their monthly cooking fuel expenditure. Thus, KOKO Networks is making a paradigm shift in access to affordable clean modern cooking fuel and immensely contributing towards SDG 7. CME will monitor the following parameter to measure this impact:

- a) Proportion of population or number of households with primary reliance on clean fuels and technology

SDG 13: The use of the KOKO Cooker and fuel for the majority of cooking significantly reduces GHG emissions through the replacement of non-renewable biomass combusted in the baseline. These GHG emission reductions are already measured, monitored and certified as CERs under the UNFCCC CDM Programme of Activities No. 10476 KOKO Kenya - Ethanol Cook stoves Program. Under this VPA we therefore only seek labeling of these CERs.

In summary, this VPA will be seeking one or more of the following Gold Standard Certified Impacts and Products:

- SDG 3 GS certified **Health Improvements Outcome Statement**
- SDG 5 GS certified **Gender Equality Outcome Statement**
- SDG 7 GS certified Access to Clean Energy Outcome Statement
- SDG 13 **GS VERs**

Project Chronology

The complete chronology of the project (VPA-2) is provided below:

Event	Date
Project Start Date	18 th July 2020 (Date of sale of 1 st cooker)
CDM Registration of the Project (Inclusion Date)	28 th August 2020
GS Listing of the Project	2 nd March 2022
Completion of GS Validation	6 th June 2022
Completion of Design Review Publication	25 th August 2022

A.4. Scale of the project

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The scale of the Gold Standard CPA is determined in accordance with the applicable Gold Standard Guidelines (Annex B – Requirements & Procedure to transition projects from other standards to Gold Standard for the Global Goals, GHG Emissions Reduction & Sequestration Product Requirement Version 2.0). As per the guidelines (section 5.1.2 & 5.1.3), CDM project shall maintain project scale as registered in accordance with CDM project standard for project activities. The scale of the project defined in the CDM registered CPA-DD is small scale in accordance with the CDM modalities and procedures. Hence, the same scale “Small Scale” is retained for GS-CPA inclusion. In accordance with the Micro-scale Additionality Tool, the rated capacity of the individual cookstove will be less than threshold limit defined for “microscale CDM units” i.e. 1500 kW. Hence the CPA will be exempted from microscale aggregate threshold and bundling requirements. CME will ensure that all CPAs are solely composed of “microscale CDM units”, hence the CME is not required to demonstrate compliance with the small-scale CDM thresholds at the aggregate level of the CPA. As a result, unlimited cookstove units can be included in a CPA under this PoA.

The CME submitted a deviation from the small-scale threshold limits as defined in section 9.1.2 of the GHG Product Requirements Standard and request the Gold Standard to conduct design review of the project by evaluating it against the guidelines of Annex B of the GHG Product requirements. The review was approved on 27th April 2022. The approved deviation (DEV_247) has been submitted on GS portal.

Gold Standard

A.5. **Funding sources of project**

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The project is funded predominantly through equity from individual investors and private equity firms forming 80% of the total capital. KOKO Networks has also raised 18% of capital as debt from commercial lenders. The remaining 2 % of capital is raised in form of grants (both repayable and non-repayable) from various non-governmental organizations (NGOs). However, none of the funding sources can be considered as diversion of Official Development Assistance or public funding of the project.

Section B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES)AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

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Approved small scale methodology AMS-I.E 'Switch from non-renewable biomass for thermal application by the user' version 09.0

<http://cdm.unfccc.int/methodologies/DB/IO5FJLJFWT91R6B8SO5BC7TXSK27I2>

Reference Tools & Methodology:

- Tool 19: Demonstration of additionality of microscale project activities, Version 08.0
- Tool 30: Calculation of the fraction of non-renewable biomass, Version 1.0
- TOOL03: Tool to calculate project or leakage CO2 emissions from fossil fuel combustion, Version 3
- TOOL16: Project and leakage emissions from biomass, Version 4
- TOOL05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, Version 3.0
- TOOL12: Project and leakage emissions from transportation of freight, Version 1.1.0

B.2. Applicability of methodology (ies)

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Applicability Condition of AMS-I.E	Justification
1. This methodology comprises of activities to displace the use of non-renewable biomass by introducing renewable energy technologies. Examples of these technologies include, but are not limited to biogas stoves, bio-ethanol stoves, solar cookers, passive solar homes	The CPA involves distribution of bio-ethanol stoves for cooking. Bioethanol in Kenya and other East African countries is procured from molasses which is an agro-waste residue and hence is classified as renewable source. Globally, about 93% of ethanol is produced from renewable sources owing to better economic feasibility². Since the smart cookers technology is solely dependent on the dedicated fuel supply through KOKOpoints (no external fuel filling possible due

² <http://www.essentialchemicalindustry.org/chemicals/ethanol.html>

	to proprietary valve system), they will only operate on bioethanol which will be centrally procured by the CME through a framework agreement with Vivo Energy (Shell Licensee), one of the most reputed Fuel Marketing Companies (FMCs)³. The fuel protocol as part of framework agreement prohibits the procurement of synthetic ethanol for this PoA. In accordance with the framework agreement, an independent analyst is collaboratively hired by CME and FMC, to test the product against the supplier's Certificate of Quality (CoQ) for conformance and verify the chain-of-custody documents establishing the fuel traceability till the bioethanol production site (synthetic ethanol requires completely different feedstock like ethylene/natural gas and hence cannot be produced in bioethanol production sites). The fuel will be approved for further distribution only if the fuel conforms to the standards as defined in the "Bioethanol Product specification" document and all chain-of custody documents are found complete. The process will ensure that only bioethanol is consumed in the PoA. The implementation of project activity will result in displacement of non-renewable fossil fuel by renewable bio-ethanol fuel in households in Kenya.
2. Project participants are able to show that non-renewable biomass has been used since 31 December 1989, using survey methods or referring to published literature, official reports or statistics	In 1980, 95% of total non-commercial energy demand in Kenya was met through fuelwood and charcoal⁴. There has not been significant change in the non-commercial energy scenario since 1980 and the majority of household energy demand was met through charcoal and fuelwood even in year 2017 in Kenya.

³ <https://www.vivoenergy.com/About/Vivo-Energy-at-a-glance>

⁴

	According the baseline analysis of the National Climate Change Action Plan 2013-2017 and Second National Communication to UNFCCC, the LULUCF sector is the second largest contributor to Kenya's GHG emissions after agriculture, accounting for 32.5% of emissions in 2015 largely as a result of deforestation through clearing of forested lands for agriculture, wood harvesting, for fuelwood, charcoal and other wood products and urban development or settlement. It can therefore be concluded that non-renewable biomass has been used in Kenya since long back before 31st December 1989.
3. The methodology is applicable for technologies displacing use of non-renewable biomass by renewable energy	The technologies implemented under this CPA will be energy efficient cook stoves that use bio-ethanol fuel from molasses, which is a renewable biomass and will displace the use of fuel wood, which is a non-renewable biomass used by the baseline stoves.
4. Project participants or coordinating and managing entities shall describe in the PDD/PoA-DD how the double counting of emission reductions has been addressed (e.g. between end users, distributors and producers of stoves)	All the activities starting from production to distribution to operation of KOKO smart cookers is being centrally managed by the CME through a digital platform connected to all users via mobile network. Each smart cooker will be tagged by a unique identification number which will be linked to the NFC chip attached to the smart cooker. This technology is exclusive to KOKO smart cookers and not available to previously registered CDM projects. Hence, there is no possibility of double counting of the cookstoves. Any double counting due to discontinuation of existing CDM project is also addressed through a robust management system which is discussed in section H.3 above.
5. For project activities introducing bioethanol cookstoves project participants or coordinating and managing entities shall demonstrate that the	The KOKO Cookstoves, to be distributed under the CPA, are designed and constructed in accordance with the national safety and performance regulation on ethanol cookstoves of Kenya – "Ethanol Fuelled Cooking Appliances

bioethanol cookstoves are designed, constructed and operated to the requirements (e.g. with regard to safety) of a relevant national or local standard or comparable literature. Latest guidelines issued by a relevant national authority or an international organisation may also be used.	Specifications” ⁵ . As an illustration, the “KOKO Cooker” stove kit consisting of a two-burner stove and a durable ‘smart’ canister, with thermal efficiency of 60% and thermal output of 2.1 kW (against the requirement of 45% efficiency and 1.4 kW output) fully comply with the above standard.
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B.3. Project boundary

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Source		GHGs	Included?	Justification/Explanation
Baseline scenario	Combustion of Non-renewable biomass/fossil fuel for cooking in households	CO ₂	YES	Main source of emissions
		CH ₄	NO	Not required by the Methodology
		N ₂ O	NO	Not required by the Methodology
Project scenario	Combustion of non-renewable bio-mass/fossil fuel for cooking in households	CO ₂	YES	Main source of emissions
		CH ₄	NO	Not required by the Methodology
		N ₂ O	NO	Not required by the Methodology
	Energy Consumption associated with Production of Feedstock or renewable fuel and its distribution	CO ₂	YES	Minor source of emissions
		CH ₄	NO	Not Applicable
		N ₂ O	NO	Not Applicable

B.4. **Establishment and description of baseline scenario**

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As per the applied methodology, the baseline scenario, in the absence of the project activity, would be the use of fossil fuels for meeting similar thermal energy needs.

Kenya, primarily being a low-income country, historically has high dependence on biomass fuel for total energy use. More than 2/3rd of the total energy requirement is met by the biomass fuel in Kenya. For household energy purpose, the dependency on biomass fuel is even higher as described below in table 1.

The Second National Communication to UNFCCC states:

"About 87 per cent of the country's domestic energy demand is met by biomass particularly wood fuel, which provides 90 per cent and 85 per cent of rural and urban households' energy requirement, respectively. The need for wood has led to substantial deforestation and land degradation. Access to modern energy services is required to reduce the wood fuel dependency."

Since economic liberalization in 1993, Kenya is enjoying a positive growth rate albeit fluctuating in magnitude. However, access and penetration of modern and clean fuel for household energy use has still been a challenge in Kenya. In last 2 decades, the use of charcoal has increased in Kenya due to many factors like convenience and migration. The production of charcoal from wood is highly inefficient and is often harvested illegally, leading to massive scale deforestation in Kenya. Over 95% of the kilns used in the country are considered inefficient, with conversion efficiencies in the range of just 8-20%.

In recent years, the Govt. of Kenya has taken cognizance of the inefficiencies in household energy sector and taking steps to mainstream modern clean fuels like LPG. However, it is facing daunting challenge in significantly altering the consumption pattern due to various reasons despite having significant health, environmental and economic benefits. Few of these reasons are lack of consumer awareness, affordability and accessibility barriers associated with LPG.

Due to low penetration of LPG, the majority of households are still using traditional cookstoves which are highly inefficient and unhealthy. As a result, Kenya loses 10.3 million m³ of wood from its forests every year from firewood and charcoal consumption, a major contributor to the country's 0.3% annual deforestation rate⁶. Therefore, the heavy dependence of households over woody biomass (fuelwood and charcoal) is evident and establishes woody biomass as the baseline fuel for cooking purpose within Kenya.

The Kenyan power sector policies and regulations

The power sector falls under the Ministry of Energy (MoE) and is regulated by the Energy Regulatory Commission (ERC). The latter regulates the entire energy sector and is empowered to set, review and adjust tariffs for sale and to ensure competition in the power sub-sector.

The energy policies and regulations of Kenya relevant to this PoA are given below:

- The Energy Act 2006: National policy and strategy document for short to long-term energy development in Kenya
- Feed-in Tariff Policy 2008: Development of electricity generation projects from renewable sources
- The Energy (Energy Management) Regulations 2012: Promotion of Energy Management and Conservation in Kenya
- The Energy Bill 2017: National Integrated Energy Policy

None of the above policies give any type of comparative advantage to a particular technology or fuel and hence cannot be classified as E+ or E- policy.

B.5. Demonstration of additionality

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The table below is only applicable if the proposed project is deemed additional, as defined by the applied approved methodology or activity requirement or product requirement.

6

⁶ "Scaling up clean cooking in urban Kenya with LPG & Bio-ethanol" by Dalberg

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	Does the VPA comply with the additionality criteria: Positive list (Annex B) article 1.1.3: 'Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in <= 600 MWh of energy savings per year or <=600 tonnes of emission reductions per year.' The 600 MWh of energy savings translated to 68.4 kW of thermal output of cookstove. [Max. Operating Hours – 8760 hours Energy Saving Threshold – 600,000 kW Max. Threshold for Thermal Capacity – $600,000/8760 = 68.5 \text{ kW}$
Describe how the proposed project meets the criteria for deemed additionality.	Yes. For inclusion in the PoA, each VPA has to demonstrate that it meets the following requirement: "Each cookstove model included in the VPA will meet the design threshold factor of 68.5 kW of thermal power" The KOKO Cooker thermal output is well below this threshold at 2.1 kW.

B.5.1. Prior Consideration

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Not Applicable, only non-CER, Retroactive projects and all projects undergoing design change required prior consideration by submission timeline. This is a UNFCCC approved CDM project and hence prior consideration not required.

B.5.2. Ongoing Financial Need

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Not Applicable

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Gold Standard

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
SDG Goal 3: Ensure healthy lives and promote well-being for all at all ages	Target 3.9: By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.	Number of KOKO Users /households experiencing reduction in the indoor air pollution in their kitchen
SDG Goal 5: Achieve gender equality and empower all women and girls	Target 5.5: Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making in political, economic and public life	Number of women in managerial positions
SDG Goal 7: Affordable and Clean Energy	Target 7.1: By 2030, ensure universal access to affordable, reliable and modern energy services	1. Proportion of population or number of households with primary reliance on clean fuels and technology
SDG Goal 13: Take urgent action to combat climate change and its impacts	Target 13.2. Integrate climate change measures into national policies, strategies and planning	Total GHG emission reduction achieved by the project

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

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SDG 3

The contribution to SDG 3 is calculated as the number of households confirming improvement in air quality. To collect information on the improvement of air quality,

the Monitoring Survey will include questions relevant to perform comparative analysis of indoor air pollution, for instance:

Survey Question	Option 1	Option 2	Option 3
Do you use KOKO Cooker Regularly?	Yes	No	Alternatively
Do you experience any difference in smoke produced in kitchen while cooking with KOKO Cooker (as compared to baseline appliance)?	No difference	More Smoke	Less Smoke

The share of households confirming less smoke by the improved stove in the Monitoring Kitchen Survey will be multiplied with the number of stoves disseminated under the specific CPA and will result in the number of households confirming improvement in air quality.

$$S_{\text{(less smoke)}} = S_{\text{install}} \times HH_{\text{(less smoke, y)}} \times U_{\text{(p, y)}}$$

$S_{\text{(less smoke)}}$ Percentage of KOKO Users with improved in-air quality

S_{install} Number of stoves installed

$HH_{\text{(less smoke, y)}}$ Share of KOKO users confirming less smoke

$U_{\text{(p, y)}}$ Usage rate in year y

SDG 5

SDG 5 statements will be claimed on the basis of female staff and management enrolments at KOKO networks which will be verifiable from the Staff HR Records.

SDG 7

SDG 7 statements will be claimed on the basis of regular operations of eligible cooker sold to customers by KOKO Networks and the local household population. This will be verifiable from the Customer database of KOKO Cookers. The operational status of the cookers will be determined based on the periodic survey conducted by the CME. The procedure for survey is discussed in section B.7.2 Sampling Plan below.

N_{HH} – Number of households using KOKO Cooker

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SDG 13

SDG 13 product “Emissions Reductions” are calculated by selecting appropriate methodological choices provisioned in the approved methodology AMS-I.E Version 9. The detailed equations, calculation approach and sample calculations are provided in the registered CDM CPA-DD for CPA-0002.

B.6.2. Data and parameters fixed ex ante

The fixed ex-ante parameters required for SDG 13 (emission reduction) calculation are reported in the CDM CPA-DD for CPA-0002 and summarized below in section B.6.3. There are no fixed/default parameters required for other SDGs as all required parameters are monitored.

B.6.3. Ex ante estimation of SDG Impact

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SDG 3

Parameter	Ex Ante Estimation	Source
S_install: Number of stoves installed	200,000	Customer Database
HH_(less smoke,y): Share of KOKO users confirming less smoke	75%	CME Conservative Assumption
U_(p,y): Usage rate in year y	99.5%	Previous CME Surveys
S_(less smoke): KOKO Users with improved in-air quality	149,250	Calculated

SDG 5

1. Number of women in managerial positions

The total number of people employed by KOKO Networks at all its offices is 450. As per the internal HR records, the share of female strength employed in KOKO Networks is 35%, which results in ~160 female employees at KOKO. Out of 160 female employees, 7 women are working in managerial positions which is included as monitoring parameter for annual monitoring. The number of females on managerial positions can be verified from the staff records maintained by KOKO HR Department.

SDG 7

Parameter	Ex Ante Estimation	Source
S_install: Number of stoves installed	200,000	Customer Database
U_(p,y): Usage rate in year y	99.5%	Previous CME Surveys
N_HH	199,000	Calculated

SDG 13

SDG 13 product “Emissions Reductions” are calculated by selecting appropriate methodological choices provisioned in the approved methodology AMS-I.E Version 9. Please refer to UNFCCC registered CDM CPA-DD for CPA-0001 for complete details of SDG 13 ex ante parameters. The following ex-ante parameters are used for estimation of SDG 13 (Emission Reduction), which are approved under CDM UNFCCC.

Parameter Name	Symbol	Value	Unit	Source
Net Calorific Value of Fuelwood	NCV _{biomass}	0.0156	TJ/Tonne	Default Value, AMS-I.E. Ver-9
Net Calorific Value of Charcoal	NCV _{charcoal}	0.0295	TJ/Tonne	Default Value, AMS-I.E. Ver-9
Charcoal Conversion Factor	CF _{Charcoal}	6.0	Unitless	Default value, AMS-I.E. Ver-9
Fraction of woody biomass saved by the project activity that can be established as non-renewable biomass	f _{NRB}	0.9268	Fraction	Re-calculated value as per CDM Tool 30
Emission factor for the substitution of non-renewable woody biomass	EF _{projected fossil fuel}	63.70	Tonnes CO ₂ e/TJ	Default Value, AMS-I.E. Ver-9

Leakage Adjustment Factor (LAF)	LAF	0.95	fraction	Default Value, AMS-I.E. Ver-9
Net Calorific Value of Ethanol	NCV _{eth}	0.027	TJ/Tonnes	IPCC default value
Density of Ethanol	D _{ethanol}	0.79	Tonnes/kL	IPCC default value
Efficiency of baseline appliance being replaced	h _{old,i}	20.00%	%	Default Value, AMS-I.E. Ver-9
Average annual consumption of woody biomass per person before the start of the project activity	BC _{BL,PP,y}	1.71	Tonnes/person/year	Registered CPA-DD-0002
Average number of persons served per household	N _{p,HH}	4.0	persons/household	Registered CPA-DD-0002
Project Emission Factor for Bioethanol Production	EF _{bioethanol_production}	8.73	g CO ₂ e/Litre of Bioethanol	Registered CPA-DD-0002
Efficiency of bioethanol KOKO Cooker	h _{Eth}	60%	%	Registered CPA-DD-0002
Emission factor for electricity generation	EF _{EF,j,y}	1.3	Tonnes CO ₂ e/MWh	Registered CPA-DD-0002
Average technical transmission and distribution losses for providing electricity	TDL _{j,y}	20%	%	Registered CPA-DD-0002
Default CO ₂ emission factor for freight transportation	EF _{CO₂,f}	245	g CO ₂ e/tonne-km	Registered CPA-DD-0002

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 3

Year	Baseline estimate	Project estimate	Net benefit
Year 1	N/A	N/A	149,250

Year 2	N/A	N/A	149,250
Year 2	N/A	N/A	149,250
Year 4	N/A	N/A	149,250
Year 5	N/A	N/A	149,250
Total			746,250

Total number of crediting years	5		
Annual average over the crediting period	N/A	N/A	149,250

SDG 5

Year	Baseline estimate	Project estimate	Net benefit
Year 1	N/A	N/A	7
Year 2	N/A	N/A	8
Year 3	N/A	N/A	15
Year 4	N/A	N/A	20
Year 5	N/A	N/A	25
Total			75

Total number of crediting years	5		
Annual average over the crediting period	N/A	N/A	15

SDG 7

Year	Baseline estimate	Project estimate	Net benefit
Year 1	N/A	N/A	199,000
Year 2	N/A	N/A	199,000
Year 3	N/A	N/A	199,000
Year 4	N/A	N/A	199,000
Year 5	N/A	N/A	199,000
Total			995,000

Total number of crediting years	5		
Annual average over the crediting period	N/A	N/A	199,000

SDG 13

Year	Baseline estimate	Project estimate	Net benefit
Year 1	1,083,190.70	1297.58	1,081,893
Year 2	1,083,190.70	1297.58	1,081,893
Year 2	1,083,190.70	1297.58	1,081,893
Year 4	1,083,190.70	1297.58	1,081,893
Year 5	1,083,190.70	1297.58	1,081,893
Total	54,15,953.50	6,487.90	5,409,465

Total number of crediting years	5
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Annual average over the crediting period	1,083,190.70	1297.58	1,081,893
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B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Data / Parameter	Di, Date of commissioning of project device i
Unit	Date
Description	Actual date of commissioning of project device
Source of data	Sales Database
Value(s) applied	[XX]
Measurement methods and procedures	Each sale will be recorded in sales database along with the name of recipient, usage type, contact details, location of household (village, district etc)
Monitoring frequency	Recorded at the time of selling of project devices
QA/QC procedures	The information will be recorded electronically through a robust registration process and archived on cloud system. Periodic backup of the complete database will be automatically taken by the system.
Purpose of data	Used for calculating StoveYear(i), fraction of year(s) the cookstove i was operational during the monitoring period which is applied for baseline emission calculation
Additional comment	StoveYear(i) = Minimum of { (End-Date of Monitoring Period – Di, Date of Commissioning of Project Device) AND (End-Date of Monitoring Period – Start-Date of Monitoring Period) } / 365

Data / Parameter	N _{HH}
Unit	Number

Description	Number of project devices in households of type i and batch j operating during year y
Source of data	Sales database and monitoring survey
Value(s) applied	Domestic Use Cookstoves, N_{HH} : [XX]
Measurement methods and procedures	CME will maintain online database of all stoves sold along with its usage (domestic). The information will be cross-checked during project monitoring surveys for the sample cookstoves. The proportion of operating cookstoves will be determined statistically through the monitoring surveys.
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	The number of operating cookstoves of each type will be determined by multiplying the total sales of cookstoves with the proportion of operating cookstoves estimated through the monitoring survey. A 95 /10 confidence / margin of error shall be achieved for the sampling
Purpose of data	Used for calculation of baseline emissions
Additional comment	N/A

Data / Parameter	$BC_{PJ,PP,y}$
Unit	Tonnes/person/year
Description	Average annual consumption of woody biomass in the household pre-project devices during the project activity, if it is found that pre-project devices were not completely displaced but continue to be used to some extent
Source of data	Project Monitoring Survey for households
Value(s) applied	Woody Biomass Consumption per household project device: [XX]
Measurement methods and procedures	The amount of charcoal and fuelwood consumed by household during a period will be separately monitored. The average value for all survey samples will be calculated.
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	The Guidelines for Sampling and Surveys will be referred to adopt the best practices for conducting the surveys.

	A 95 /10 confidence / margin of error shall be achieved for the sampling parameter irrespective of annual / biennial monitoring frequency as per para 22 of Standard: Sampling and surveys for CDM project activities and programmes of activities, Version 08.0.
Purpose of data	Used for calculation of baseline emissions
Additional comment	N/A

Data / Parameter	$Q_{HH,Eth}$
Unit	Litres/day
Description	Average daily consumption of bioethanol in a project cookstove (KOKO cooker) distributed to Households
Source of data	Project Monitoring Survey for Households
Value(s) applied	Average Daily consumption of bioethanol in project households: [XX]
Measurement methods and procedures	The average daily consumption of bioethanol is calculated from the mean monthly expenditure of household which is determined through the project monitoring survey for households.
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	The Guidelines for Sampling and Surveys will be referred to adopt the best practices for conducting the surveys. A 95 /10 confidence / margin of error shall be achieved for the sampling parameter irrespective of annual / biennial monitoring frequency as per para 22 of Standard: Sampling and surveys for CDM project activities and programmes of activities, Version 08.0.
Purpose of data	Used for project emission calculation
Additional comment	N/A

Data / Parameter	$N_{KP,y}$
Unit	Number
Description	Number of KOKOpoints operating during year y
Source of data	Database of installed KOKOpoints under operations
Value(s) applied	[XX]

Measurement methods and procedures	A real-time database will be maintained to keep record of all operational KOKOpoints. All KOKOpoints will be connected to internet and hence will be monitored for operational status online.
Monitoring frequency	Continuously monitored, recorded annually
QA/QC procedures	As the database will be generated from online monitoring, it will be highly reliable and does not require further review. The technical support team will ensure that each of the non-operational KOKOpoint is attended in a timely manner. Any discrepancy in the actual status and database will be highlighted by the technical team at the earliest.
Purpose of data	Calculation of project emissions due to electricity consumption at KOKOpoints.
Additional comment	No Comments

Data / Parameter	$NCV_{i,biomass}$
Unit	TJ/tonne
Description	Net calorific value of the fuel type "i" used in project scenario including non-renewable woody biomass, charcoal or renewable bio-ethanol.
Source of data	IPCC Guidelines for National Greenhouse Gas Inventories 2006 https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf
Value(s) applied	0.0156
Measurement methods and procedures	Default value for all fuels utilized in the PoA will be sourced from IPCC Guidelines for National Greenhouse Gas Inventories 2006.
Monitoring frequency	The monitoring of NCV of bioethanol (NCV_{Eth}) will be conducted either in-house using the bomb-calorimeter or by third party testing agency.
QA/QC procedures	Yearly
Purpose of data	CME will ensure that the NCV determination is performed as per the national/international standards and procedures.
Additional comment	Calculation of Project Emissions

Data / Parameter	$EC_{PJ,j,y}$
Unit	MWh/year
Description	Quantity of electricity consumed by the project electricity consumption source j in year y
Source of data	Calculated from monitoring of installed KOKOpints and number of hours during the monitoring period
Value(s) applied	[ZZ]
Measurement methods and procedures	Since the direct monitoring through meters is not possible due to decentralized units with no separate electricity meters, the overall electricity consumption of KOKOpints will be estimated by multiplying the total number of hours in the monitoring period, the rated electrical consumption of the KOKOpints (kW) and the number of KOKOpints installed (Monitored parameter). This approach assumes continuous running of KOKOpints at full capacity during the monitoring period. This is the most conservative estimation possible of electricity consumption by KOKOpints because the KOKOpint stores are only open for approximately 10 hours in a day with an average running of fuelATMs of about 1-2 hours.
Monitoring frequency	Continuous (monitoring of installed KOKOpints)
QA/QC procedures	N/A
Purpose of data	Calculation of Project Emissions due to electricity consumption
Additional comment	The power consumption rating of these KOKOpints is 20 W which results in project emissions of less than 1% of emission reductions due to electricity consumption in a given period.

Data / Parameter	$D_{f,m}$
Unit	Kilometre
Description	Return trip distance between the origin and destination of freight transportation activity f in monitoring period m
Source of data	Records of vehicle operator or records by project participants

Value(s) applied	[XX]
Measurement methods and procedures	Determined once for each freight transportation activity f for a reference trip using the vehicle odometer or any other appropriate sources (e.g. on-line sources)
Monitoring frequency	Continuous monitoring
QA/QC procedures	Wherever possible, pre-defined routes will be identified, and the actual distance of the trip will be determined through a pilot trip. The same distance can be used as default distance for all subsequent trips on the same route. In case, if there is no fixed route followed, then the actual distance of such trip will be monitored using the vehicle odometer/trip meter or any other appropriate means. Daily trips made by the vehicles were monitored for the distance travelled by either resetting the trip meter to 0 or taking the initial and final readings of the mileage meter.
Purpose of data	Calculation of project emissions due to transportation
Additional comment	No Comments

Data / Parameter	$FR_{f,m}$
Unit	Tonnes
Description	Total mass of freight transported in freight transportation activity f in monitoring period m
Source of data	Records by project participants or records by truck operators
Value(s) applied	[XX]
Measurement methods and procedures	The parameter will be recorded in accordance with tool 12 "Methodological tool: Project and leakage emissions from transportation of freight". The micro-tankers, technically designed for safely carrying ethanol, are used for transport of bioethanol from local fuel stations to the nearby KOKOpoints. A micro-tankers is completely filled up to its technical capacity before being dispatched from fuel station for KOKOpoint refills. Hence, capacity of the micro-tanker is recorded for monitoring of total mass of bioethanol transported during a monitoring period.

Monitoring frequency	Continuous monitoring
QA/QC procedures	The capacity of each micro-tanker will be recorded electronically before dispatch from local fuel stations. The record will be maintained by the CME as the local logistics are under its control.
Purpose of data	Calculation of project emissions due to transportation
Additional comment	No Comments

SDG 3

Data / Parameter	Air Quality
Unit	Qualitative (%)
Description	Users' perception on smoke reduction
Source of data	Sampling Surveys/ Annual usage survey/Monitoring survey
Value(s) applied	To be monitored
Measurement methods and procedures	Air quality will be assessed through users interviews during the User Survey. Observations as to inside/outside cooking area to confirm answers. During the survey HH users will be interviewed through direct/indirect questions to determine status of smoke reduction in the household premises.
Monitoring frequency	Annually
QA/QC procedures	N/A
Purpose of data	Sustainable Development Assessment.
Additional comment	-

SDG 5

Data / Parameter	Number of women employed by KOKO Networks Limited in managerial positions, $N_{f,KOKO}$
Unit	Number
Description	Number of women employed in managerial positions
Source of data	HR Records of the employed staff in all the registered companies owned/operated by KOKO Networks Limited
Value(s) applied	To be monitored

Measurement methods and procedures	All women managers employed by KOKO Networks (group of companies) for more than 6 months period will be accounted for this calculation.
Monitoring frequency	Annually
QA/QC procedures	N/A
Purpose of data	Sustainable Development Assessment.
Additional comment	N/A

SDG 7

Data / Parameter	Access to affordable and improved energy services, N_HH
Unit	Numbers
Description	Number of KOKO Cookers operational under the project
Source of data	Commissioning records
Value(s) applied	To be monitored
Measurement methods and procedures	Sample survey to confirm if project ICS are operational. Operational status will confirm that the users are accessed to affordable and improved energy
Monitoring frequency	Annually
QA/QC procedures	N/A
Purpose of data	Sustainable Development Assessment.
Additional comment	NA

B.7.2. Sampling plan

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The CPA shall follow the following sampling plan:

Sampling Plan

Representative sampling will be undertaken as part of a single CPA, multiple-CPA or PoA-level sampling plan that is designed in line with the requirements of the methodology applied and the Standard: Sampling and surveys for CDM project activities and programmes of activities, version 09.0.

Sample sizes will be sufficient to ensure that the precision of the sample means/ proportions are in accordance with the methodological / standard requirements. In cases where survey results indicate that desired precision is not achieved, the lower bound value of corresponding confidence interval of the parameter value may be used as an alternative to repeat the survey. Alternatively, the survey may be expanded to reach the required confidence/precision.

In order to ensure that parameter values obtained through sampling are non-biased and that data collection minimizes non-sampling (non-random, systematic) errors, the CPA will observe sound practices in designing samples and administering surveys and field measurements as outlined in the "Guidelines for sampling and surveys for CDM project activities and programme of activities" and estimation of sample size will be guided by the type of parameter being sampled that it either proportion parameter or mean parameter. The CME may choose to conduct different sampling and surveys for monitoring parameters related to different SDG impacts. For instance, CME may conduct a separate survey for SDG 13 parameters while another survey may be conducted to cover monitoring parameters of SDG 3 impacts.

The Standard (paragraph 22) mandates application of 95/10 confidence/precision for CPAs solely composed of micro-scale CDM units hence the same shall be applied.

Sampling Methodology

Sampling Objective – The sampling objective for each parameter is to determine, via sampling survey / test, a statistically significant parameter value for the emission reduction calculations and other SDG impacts. These parameters are as listed B.7.1 above and the CDM registered CPA-DD.

Field Measurement Objective and data to be collected – The value of monitoring parameters included in section B.7.1 and CDM-registered CPA-DD will be measured through appropriate statistical sampling methods. The below table summarizes the list of parameters, type and the sampling methods to be used during project surveys.

S.No.	Monitoring Parameter	Parameter Type	Target Population
1	Number of project devices distributed in households operating during year y , N_{HH}	Proportion	Cookstoves distributed in households as part of project activity
2	Number of project devices distributed in institutions operating during year y , $N_{I,i}$	Proportion	Cookstoves distributed in institutions as part of project activity
3	Average annual consumption of woody biomass per household in the pre-project devices during the project activity, if it is found that pre-project devices were not completely displaced but continue to be used to some extent, $BC_{PJ,HH,Y}$	Mean	Cookstoves distributed in households as part of project activity
7.	Air Quality	Mean	Cookstoves distributed to households
8.	Gender of User	Male/Female	Gender of the primary

			operator of the KOKO Cooker
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Target population and sampling frame – The target population is the total population served under the CPA/PoA and the sampling frame consists of aggregated data of end-users of the cookstoves as recorded in the CPA Database(s). The sampling frame will be kept for 2 years following the crediting period or the last issuance of the CERs of the project activity.

Sample method –Sampling will be conducted using either simple or stratified random sampling techniques over the sampling frame. The survey implementation and calculations will be performed as per CDM guidelines “Sampling and surveys for CDM project activities and programmes of activities”. Optionally, other sampling approaches may be used in accordance with Standard “Sampling and surveys for CDM project activities and programmes of activities” and Guideline for Sampling and Surveys for CDM Project Activities and Programme of Activities, when sampling techniques or statistical analysis necessitates it.

Desired precision / expected variance and sample size – unless otherwise noted in the measurement methods and procedures section of the monitored parameter table in section B.7.1, and as allowed by applicable methodology, the sample size will be chosen for a 95/10 precision (95% confidence interval and 10% margin of error).

Depending on the sampling type, appropriate formula for calculating sample size will be applied as described in Guideline: Sampling and Surveys for CDM project activities and programme of activities, Version 4.0. For illustration, the following formulae may be used for proportion parameters based on the sampling type:

For simple random sampling:

$$n \geq 1.96^2 * N * p(1-p) / \{ (N-1) * 0.1^2 * p^2 + 1.96^2 * p * (1-p) \}$$

Where:

n = Sample size

N = Total number of households

p = Expected proportion value

1.96 = Represents the 95% confidence required

0.1 = Represents the 10% relative precision

For Stratified random sampling:

$$n \geq 1.96^2 * N * V / \{ (N-1) * 0.1^2 + 1.96^2 * V \}$$

$$V = SD^2 / p_bar^2$$

Where:

SD² = Overall weighted-average Variance

p_bar = Overall weighted average proportion

n = Sample size

N = Total number of households

V = Expected proportion value

1.96 = Represents the 95% confidence required

0.1 = Represents the 10% relative precision

During sampling, there may be non-response from the target population. Over-sampling may be used to avoid non-response, however, sampling may be cease once required confidence/precision is met. In the case of parameters monitored for the first time the expected variation for that measure in the sample may be based on results from similar studies, pilot studies, or from the project planner's own knowledge / experience of the data (as per para 12 of Standard).

B.7.3. Other elements of monitoring plan

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To reduce monitoring efforts, a single sample set is drawn based on which various parameters shall be monitored. The CME will determine the number of samples to be monitored for each of the parameters separately. The CME may decide to stop monitoring of a parameter during the campaign once the required precision for this parameter is achieved. The monitoring team may continue to monitor appliances in

the sample with respect to the remaining parameter(s) until again the required precision for these parameters is achieved. The design of the survey questionnaire will ensure that the questions are non-intrusive and easy to understand for both the interviewee and interviewer.

Training will ensure that all monitoring staff has the appropriate skills and experience to administer relevant surveys / tests and quality checks will ensure the integrity of information flow to the CME. The CME shall review the efficacy of information gathering techniques and information flow and assess enumerator and partner feedback to make improvements as deemed necessary.

Any third parties hired by the CME, if any, to carry out sampling should have requisite skills and appropriate experience with data entry and data management. The CME will ensure that contractors are adequately trained for the tasks they are contracted for. Training will also be provided on how to deal with non-responses, refusals and other problems should these occur.

Section C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project

>> 18/07/2020 (date of 1st cooker distribution)

C.1.2. Expected operational lifetime of project

>> 10 years

C.2. Crediting period of project

C.2.1. Start date of crediting period

>> 28/08/2020

C.2.2. Total length of crediting period

>> 10 years (To be Renewed every 5 years as per GS guidelines)

Section D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. **Safeguarding Principles that will be monitored**

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 7.1 Emissions	Monitoring of Co2 emission will be done annually.
Principle 6.1 Labour Rights	Annual HSE Audits to ensure occupational safety of workers

D.2. **Assessment that project complies with GS4GG Gender Sensitive requirements**

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project respects the key gender issues and requirements of gender-sensitive design and implementation of the project. SDG#3 is one of the impact areas of the project. The project is aimed to replace conventional cook stoves with modern cook stoves. This will result in reducing use of charcoal/wood consumption.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The project does not involve in any form of based on gender, race, religion, sexual orientation or any other basis as per the country policies, strategies and best practices. The Government of Kenya reaffirms its commitment to work for the realization of constitutional guarantee of equality, social justice and non-discrimination on the basis of sex, caste, community, language and religion. The project positively contributes towards the

	gender empowerment through improvement of health and quality of life.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No gender expert will be required for the Gender Safeguarding Principles & Requirements, as this requirement is not mandatory. However, CME has implemented policies and mechanisms to align with the Gender Safeguarding Principles and Requirements of Gold Standard.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No gender experts will be present at the Stakeholder Consultation, as this requirement is not mandatory. However, the KOKO management and KOKO climate team have extensive experience in managing gender issues and conducting gender-sensitive consultations. The local stakeholder consultation was carried out in accordance with the Gender safeguarding guidelines of the Gold Standard Stakeholder Consultation and Engagement Requirements. NGOs working in Gender upliftment and rights also attended the local stakeholder consultation. Hence, Gender issues were appropriately addressed in the local stakeholder consultation meeting.

Section E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

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The PoA is registered under CDM mechanism and underwent local stakeholder consultation as per CDM requirements on 20th December 2018. The details of the consultation is provided in the CDM registered PoA-DD. The CME covered the GS stakeholder engagement requirements during the stakeholder feedback round conducted for GS-10884 & GS-10885 via online consultation, user interviews and feedback. The stakeholder feedback round was conducted as per the guidelines for retroactive GS project. Since this CPA (GS-11440) is exactly similar to GS-10885 in terms of design, geographical location (Greater Nairobi Region), implementation and operations, hence the same SFR is applicable for this CPA as well for Nairobi Network.

CME conducted another PoA-level combined stakeholder consultation and stakeholder feedback round (physical meeting) in Mombasa, Kenya which has been recently launched in November 2021. This was carried out to involve those stakeholders who were affected by the project expansion to Mombasa. The details are available in the stakeholder consultation report.

The above consultation completed by the CME meets the minimum requirement prescribed by the relevant standards. VPA02 is currently operational in Nairobi and Mombasa. The consultations conducted by CME adequately covers the minimum requirements for stakeholder engagement stipulated by GS guidelines.

All mitigation measures identified as any stage of consultation (if any) are implemented through the PoA. The details of the summary of stakeholder consultation (both initial and SFR) are provided in the stakeholder consultation report.

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
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Continuous Input / Grievance Expression Process Book (mandatory)	The grievance expression process book or the complaint book is available at the Primary Facility in Baba Dogo, Nairobi and Mombasa County.
GS Contact (mandatory)	help@goldstandard.org
Other	There is a completely dedicated centralized customer care department run by KOKO to address complaints from users and other stakeholders which is available on phone and email. A stakeholder can contact KOKO representative via free toll number – 0800 720716, free SMS on 20531 or whatsapp message on 0711900900. A strong 100+ staffed customer care department at KOKO plays a front-line role in resolving customer issues and ensuring customer satisfaction by addressing all customer queries, complaints and grievances. The key roles and responsibilities of KOKO customer care team are provided below: <u>Role of Customer Care Team</u> • Answer inbound phone calls from current and prospective KOKO customers and agents • Place outbound phone calls to current customers to solicit feedback and provide support • Conduct in-person field visits to current customer to deliver the highest quality of support • Address customer questions and concerns to ensure satisfactory issue resolution • Give voice to KOKO customers by understanding and reporting recurring questions and issues • Communicate openly and honestly with KOKO customers • Build, manage, and maintain strong, positive customer relationships • Follow existing KOKO Customer Care processes and escalate customer queries and issues to management, as appropriate The local stakeholders can also access the KOKOagents to register their comments or complaints. Complaints can also be submitted electronically to the project manager, Ed Agnew on the email E.Agnew@kokonetworks.com

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights	NO	Not relevant 1. The constitution of host country (Kenya) considers it a legal offence to violate human rights during any business activity. Kenya endorses the United Nations Guiding Principles (UNGPs) on Business and Human Rights adopted in the UN Human Rights Council (UNHRC) in 2011.	

Gold Standard

2. The Project shall not discriminate with regards to participation and inclusion		The CME complies with the legal requirements of the host country. This is not violated at any point during the project. 2. The targeted segment for facilitating clean cooking under this project is the low-income households in Kenya. The economically backward and marginalized, who form a majority of these households, will witness enhanced opportunities for participation in this programme, as this project by scaling up will reach areas where clean cooking solutions were previously unaffordable.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead	No	Not relevant	

Gold Standard

to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)		1. No, the project doesn't directly or indirectly lead to/contribute to adverse impacts on gender equality. In contrast, the project will contribute to health and well-being of women. Moreover, the project will deploy adequate measures to foster social status of women. 2. The project shall embrace the spirit of the Kenyan Employment Act and ILO guidelines. The project does apply the principles of non-discrimination and equal treatment in the production, distribution and the usage of the products. The CME also ensures equal pay for	
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		equal work during any activity in the project. 3. The project developer abides by the National Policy on Gender and Development enshrined in the Constitution of Kenya. It has a powerful mandate for equality and rights of women. Kenya has endorsed the SDG 2030 Agenda as well. The project developer recognizes and respects these commitments in the course of the project as well. There are no specific gender risks identified during the project design. 4. Not required.	
Principle 3. Community Health, Safety and Working Conditions			

1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	Not relevant Dissemination activity under the project doesn't involve any activity that triggers safety requirements. On the contrary the project is expected to alleviate existing health risks due to indoor pollution.	
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	Not relevant The project units will be simple and small in dimension. The project will not result in any change in cooking habits because of substitution of fuel. Therefore, the result of this project won't damage or remove cultural heritage, since the project is implemented in the households of families.	
>>			

Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	Not relevant The project does not involve displacement of people.	
>>			
Principle 4.3 Land Tenure and Other Rights			
a.Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership? b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or landownership?	No	Not relevant. The project does not involve any land tenure arrangements and/or other rights.	
>>			
Principle 4.4 - Indigenous people			

Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	Not relevant. The project does not involve any land/territory claim. Access to affordable modern technology will only benefit the indigenous people in improving their quality of life.	
>>			
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	Not Relevant Corruption at individual or organizational level is prohibited under the laws of Kenya. CME abides by the Bribery Act of Kenya, which prohibits the organization and its representatives to involve in corruption.	
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national	No	Not relevant 1. Kenya has ratified 7 out of the 10 fundamental ILO	

labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND		conventions including relating to abolition of forced labour, equal remuneration and no discrimination between men and women in employment and occupation. 2. The Labour Relations Act 2007 ensures the basic rights of workers are met. The project developer and its partner comply with the national labour regulations and guidelines. All the health and safety related laws are abided by in the production and distribution process. Hence, this safeguarding principle is not triggered. 3. The working conditions of the employees and workers are in compliance with the Occupational and Safety Act of Kenya. Hence, this safeguarding	
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e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures		principle will not be triggered. 4. The project does not employ and is not complicit in any form of child labour in any way. Plants are built & maintained with the help of trained staffs and not child labour. 5. Occupational Safety is paramount for the CME and deeply ingrained in the ethos of the company. Utmost safety measures are undertaken while manufacturing the products to be disseminated under the project. Hence the safeguarding principle is not triggered.	
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Principle 6.2 Negative Economic Consequences			
1.	Does the project cause negative economic consequences during and after project implementation?	No	The project design ensures financial sustainability by a combination of revenue sources (majorly from carbon credit sales and revenue from cooker sales). It is expected that as the users have already benefited the economic savings by using the fuel efficient stoves, majority of them would purchase these again even after the certification period. The project does not possess any threat to the local economy, instead it catalyses several aspects relating to it's growth (eg., monetary savings on fuel, job creation, etc)
>>			
Principle 7.1 Emissions			

Will the Project increase greenhouse gas emissions over the Baseline Scenario?	NO	The project will deploy efficient cooking stoves against the traditional cooking stoves will replace traditional methods which will ultimately result in GHG emissions reduction.	
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project will deploy efficient cooking stoves that results in reduced consumption of firewood and charcoal. The bioethanol supplied as fuel for KOKO cookers is produced from molasses which is a low value agro residue of the sugarcane crop. Hence, it would not lead to any concerns related to energy supply.	
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			

Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project doesn't involve any activity related to discharge of waste-water, extraction of surface or ground water. This would not involve affecting the existing watercourses, watersheds or flow variability. Hence, this safeguarding principle will not be triggered.	
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion? b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?	No	The project doesn't involve any activity related to discharge of waste-water, extraction of surface or ground water. This would not involve affecting the existing watercourses, watersheds or flow variability. Hence, this safeguarding principle will not be triggered.	
>>			

Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The bioethanol supplied as fuel for KOKO cookers is produced from molasses which is a low value agro residue of the sugarcane crop. Hence, it would not lead to any concerns related to energy supply. Hence, it will not lead to any modification of the cropping pattern or landscape.	
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The project does not involve any fixed large scale installation or civil construction. Hence, it will have no adverse impact on the natural disaster risk.	
>>			

Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	The project does not involve any GMO process or technology.	
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	Not relevant. The project only has positive environmental impact via prevention of deforestation and reduction in indoor air quality.	
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			

Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	Not relevant. No hazardous waste is released due to the project operations.	
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	Not relevant. The project is not an agriculture project.	
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	Not relevant. The project does not involve harvesting of forests or its products.	
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality	No	Not relevant.	

of food available such as through crop regime alteration or export or economic incentives?		The project does not involve modification of crop regime. The project does not alter the process of cooking and hence will not impact the nutritional level of the cooked food.	
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	Not relevant. The project does not involve animal husbandry.	
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	Not relevant. The Project does not physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified	
>>			

Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?	NO	Not relevant. The project boundary is restricted to households and will not be covering any geographical area which forms a natural habitat of the endangered species. The project does not involve any large scale civil or electrical installation or modification of any land use. Hence, it will not have any impact on the endangered species within the host country.	
>>			

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	KOKO Networks Limited
Registration number with relevant authority	N.A.
Street/P.O. Box	Westlands
Building	Level 5, Rivaan Centre
City	Nairobi
State/Region	Nairobi
Postcode	N.A.
Country	Kenya
Telephone	+254 718 472 200
E-mail	g.murray@kokonetworks.com
Website	www.kokonetworks.com
Contact person	Greg Murray
Title	CEO
Salutation	Mr
Last name	Murray
Middle name	N/A
First name	Greg
Department	N/A
Mobile	+254 718 472 200

Direct tel.	N/A
Personal e-mail	g.murray@kokonetworks.com

APPENDIX 3-SUMMARY OF APPROVED DESIGN CHANGES

Not Applicable

Revision History

Version	Date	Remarks
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption